

Shaliniv54 / ISEA-Bridging_2026

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86a928f · 7 hours ago 🕒

📁 Lab 1	Add files via upload	19 hours ago
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ISEA Bridging 2026 – Lab Progress

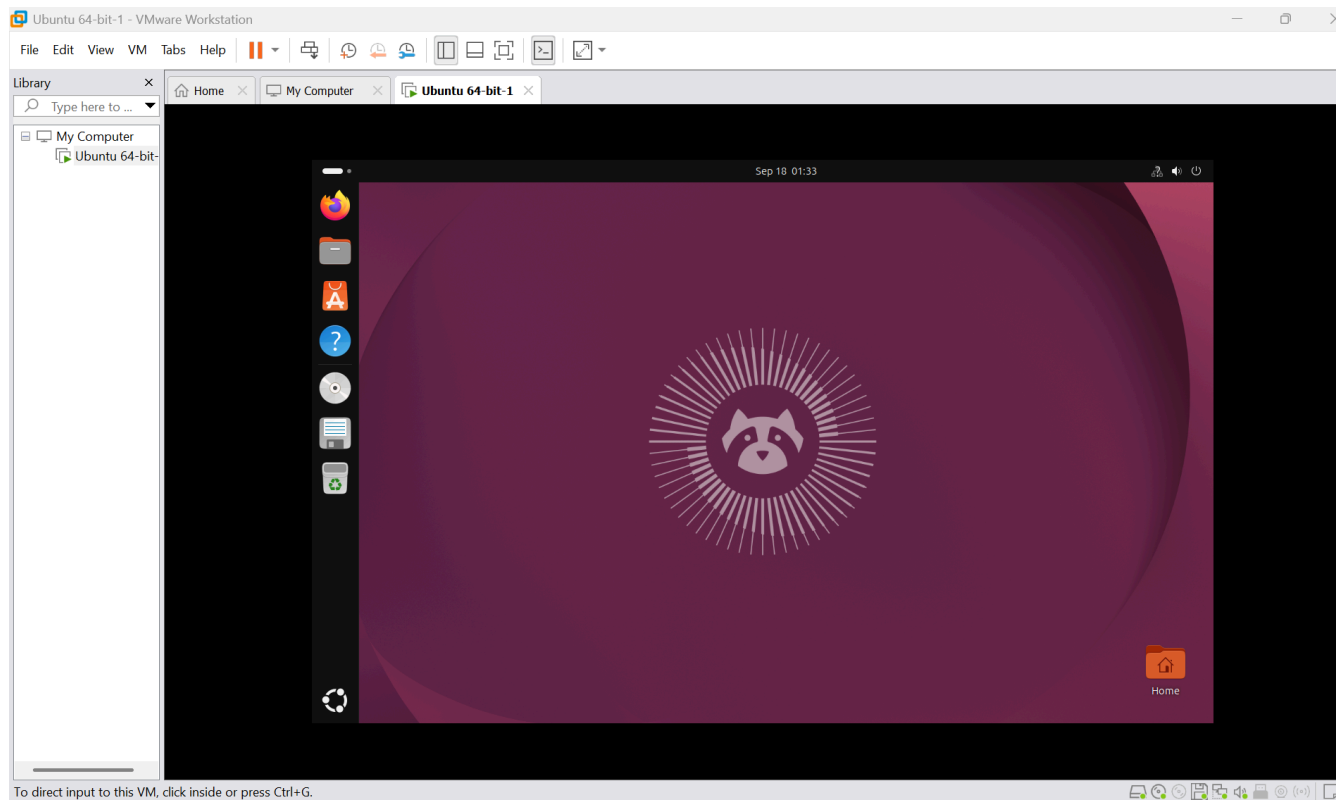
Introduction to Server Environments and Architectures (Kaplan / Murdoch). Environment: VMware Workstation, Ubuntu VM. AWS EC2 (Singapore) and Azure for Students for cloud labs.

Progress

Lab	Topic	Status
1a-1	Virtualisation & Linux setup	Done
1a-2	CLI familiarisation	Done
1b	Linux services – Apache	Done
2a	TCO comparison	Done
2b	Cloud VM + bash scripting	Done
3b	Scripting & cron jobs	Done
4a	Additional server service (MariaDB)	Done
3a	DNS & HTTPS certificates	To do

Lab 1a-1 – Virtualisation & Linux setup

Installed Ubuntu in VMware Workstation and booted to the desktop.

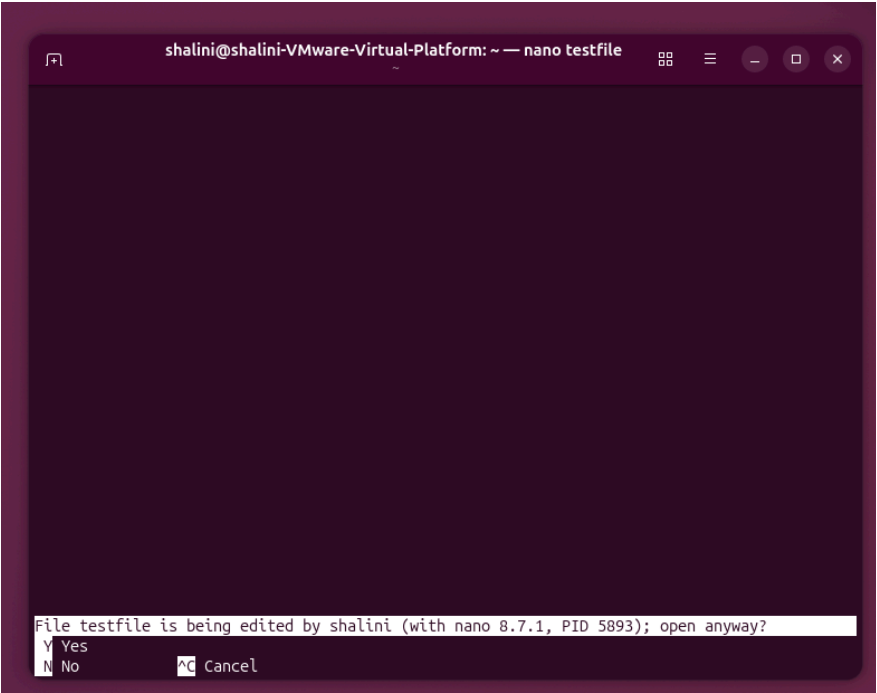
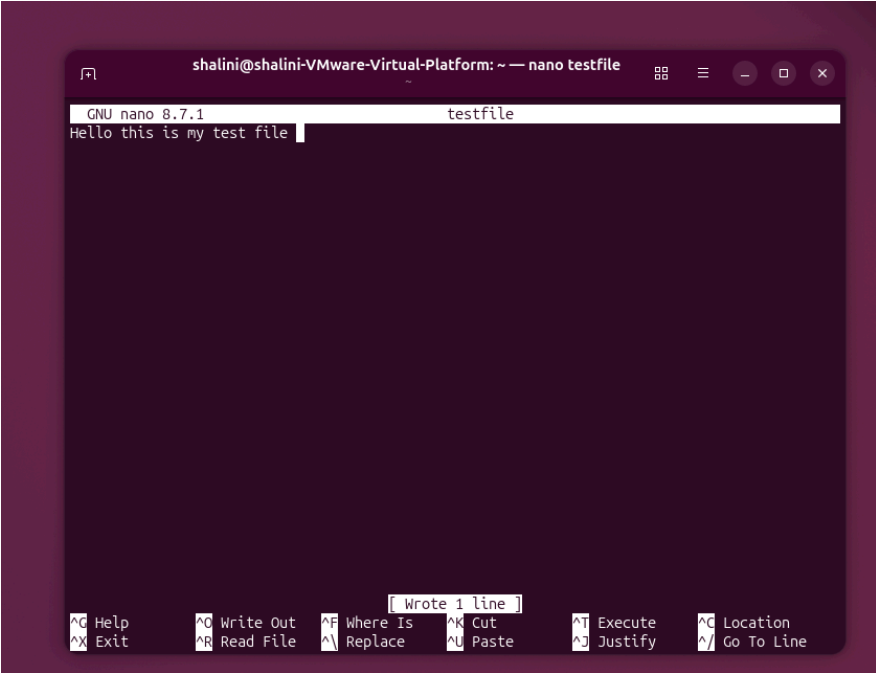


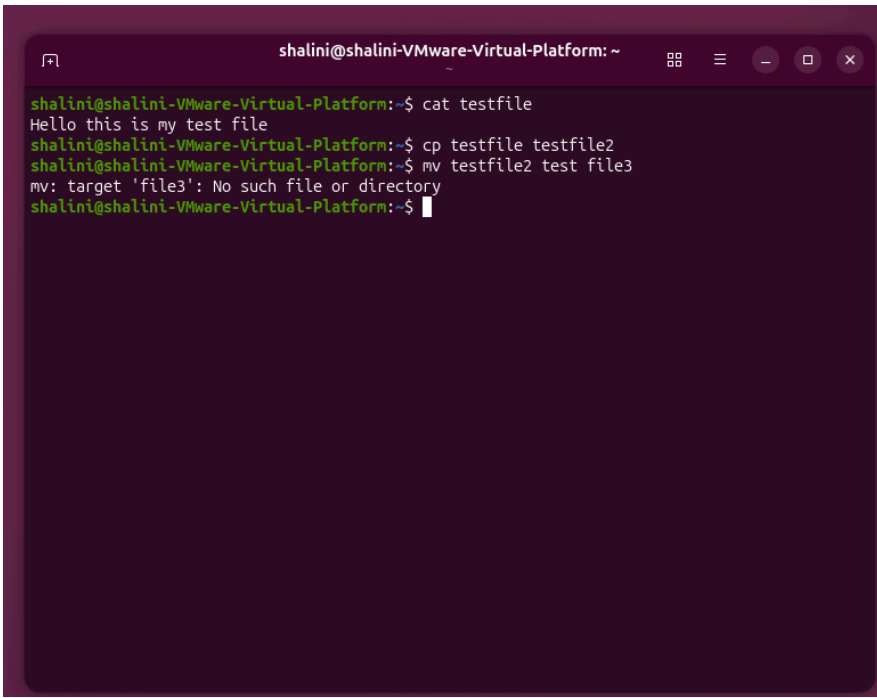
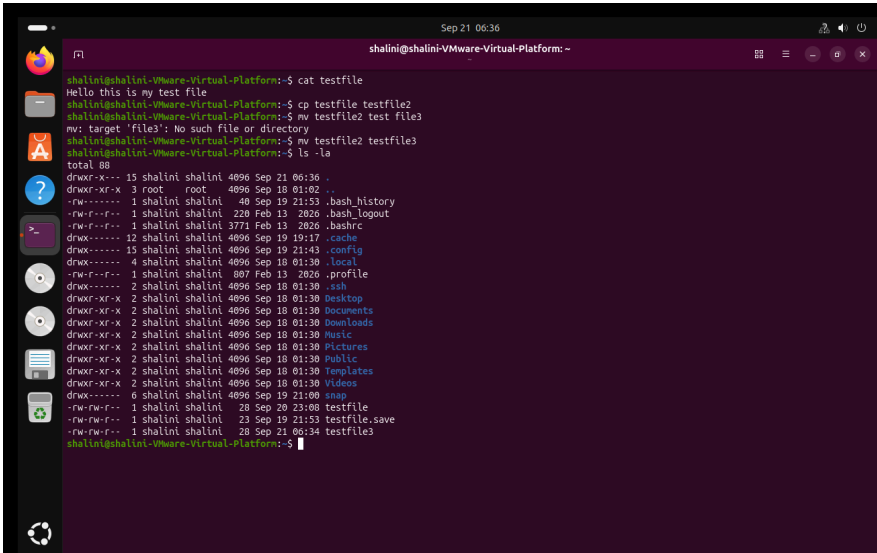
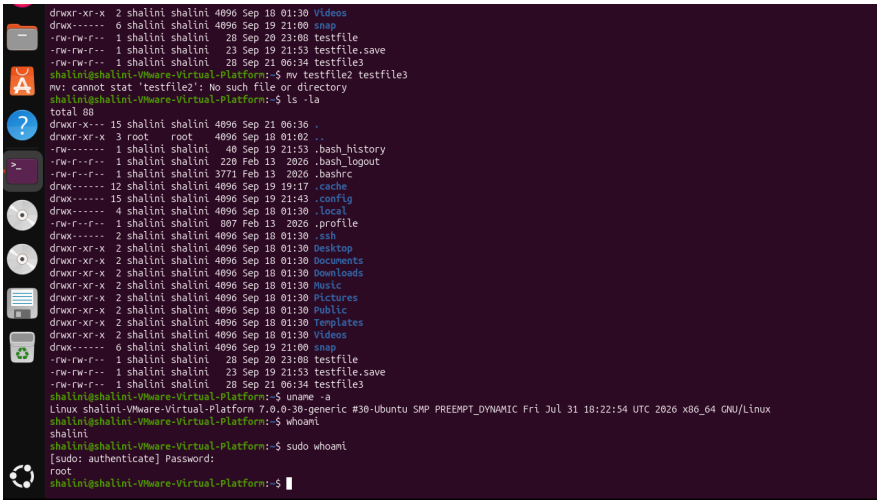
Issue: [add: password lockout, VM rebuild, what you learned]

Lab 1a-2 – CLI familiarisation

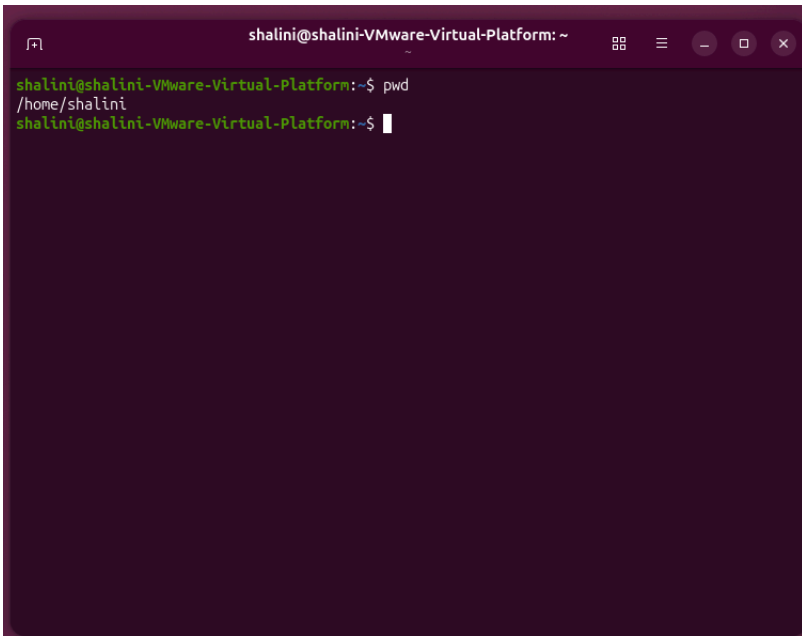
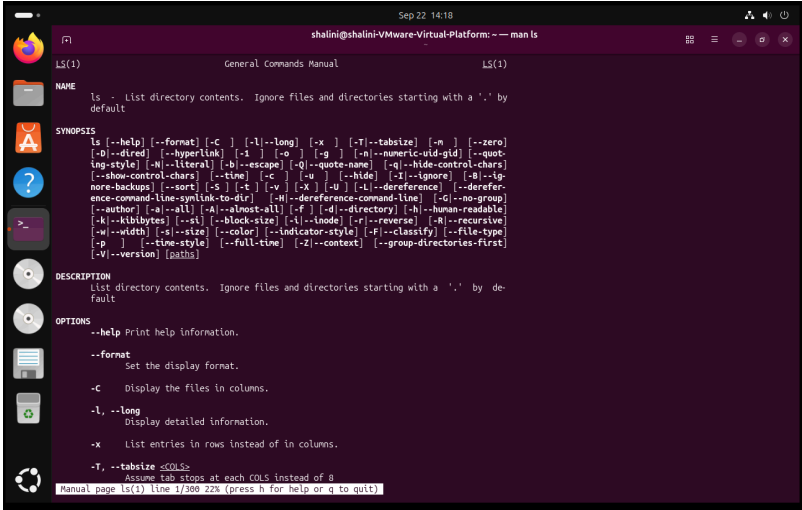
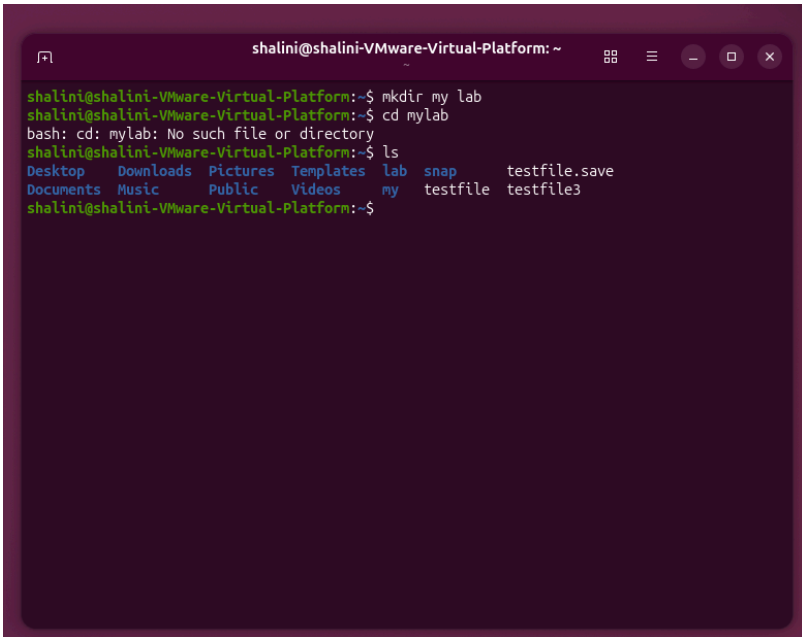
Commands used: touch , nano , cat , cp , mv , ls -la , uname -a , whoami , sudo .

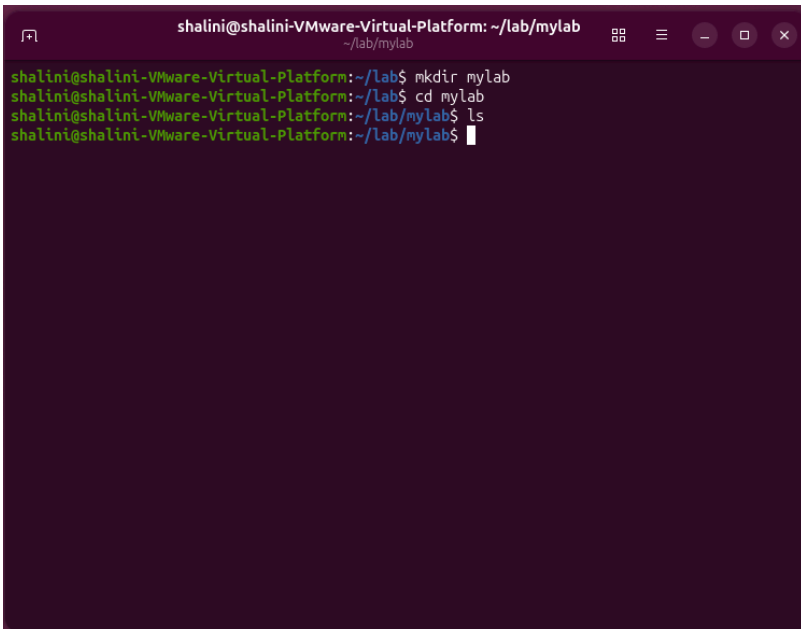
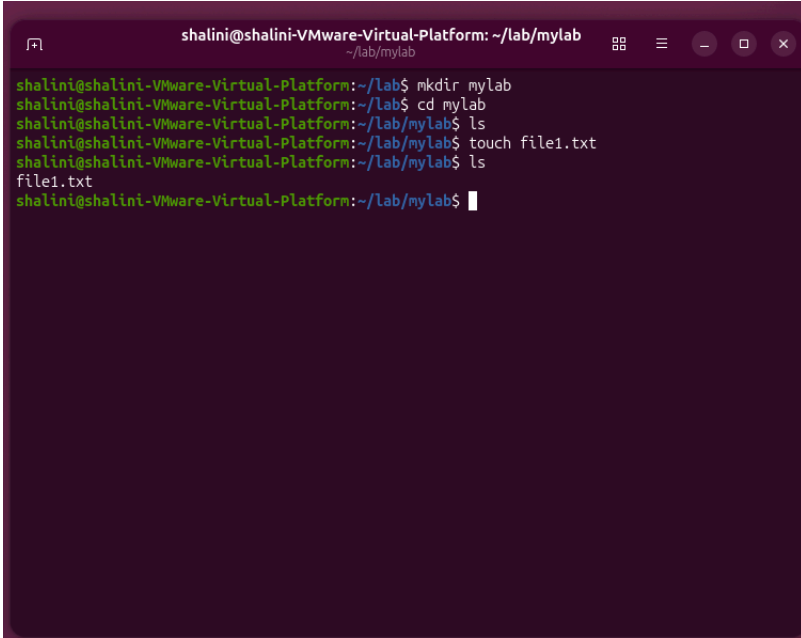
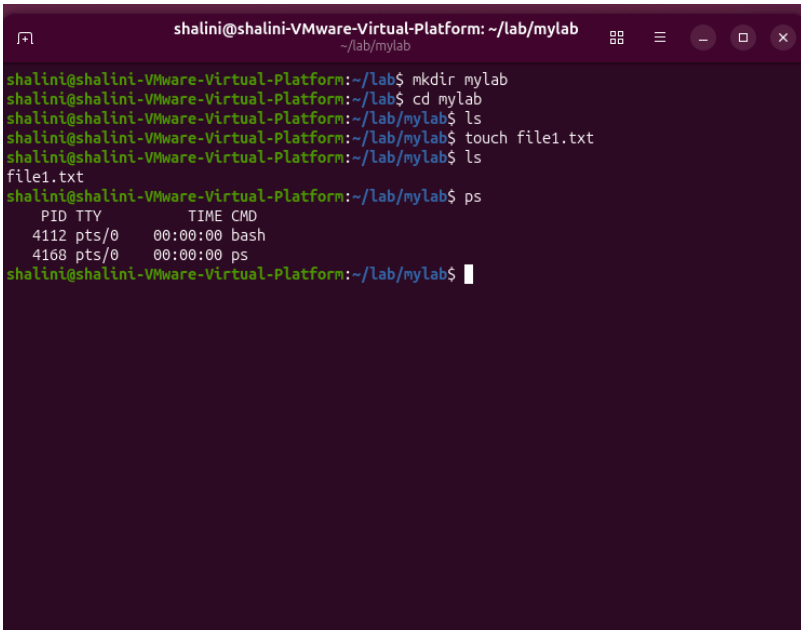
Step	Screenshot
Create file and open in nano	A screenshot of a terminal window titled "shalini@shalini-VMware-Virtual-Platform: ~". The terminal shows two commands being executed: "touch testfile" and "nano testfile". The prompt "shalini@shalini-VMware-Virtual-Platform:~\$" is visible before each command.

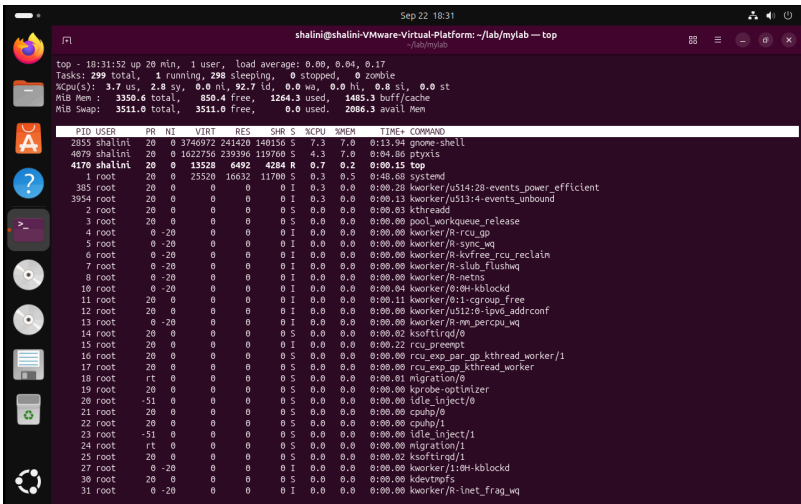
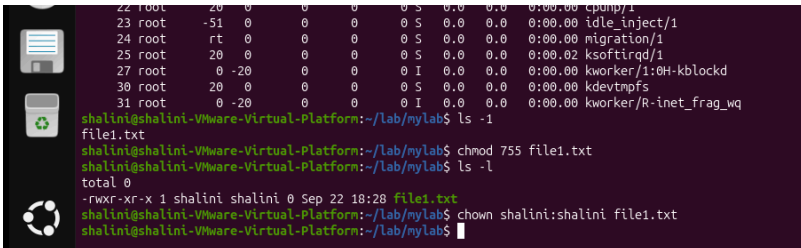
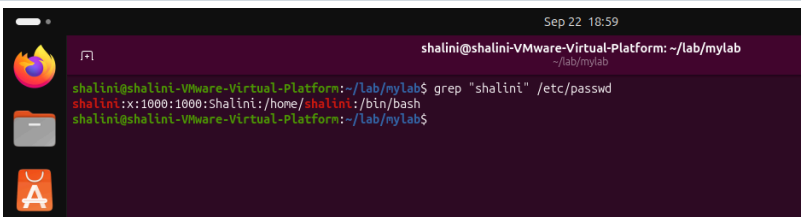
Step	Screenshot
nano warned the file was already open in another session	
Typed content and saved	

Step	Screenshot
cat , cp , then a mv typo (test file3 instead of testfile2) gave an error	 <pre> shalini@shalini-VMware-Virtual-Platform: ~ shalini@shalini-VMware-Virtual-Platform:~\$ cat testfile Hello this is my test file shalini@shalini-VMware-Virtual-Platform:~\$ cp testfile testfile2 shalini@shalini-VMware-Virtual-Platform:~\$ mv testfile2 test file3 mv: target 'file3': No such file or directory shalini@shalini-VMware-Virtual-Platform:~\$ </pre>
Corrected mv , confirmed with ls -la	 <pre> shalini@shalini-VMware-Virtual-Platform: ~ shalini@shalini-VMware-Virtual-Platform:~\$ cat testfile Hello this is my test file shalini@shalini-VMware-Virtual-Platform:~\$ cp testfile testfile2 shalini@shalini-VMware-Virtual-Platform:~\$ mv testfile2 test file3 mv: target 'file3': No such file or directory shalini@shalini-VMware-Virtual-Platform:~\$ mv testfile2 testfile3 shalini@shalini-VMware-Virtual-Platform:~\$ ls -la total 88 drwxr-xr-x 15 shalini shalini 4096 Sep 21 06:36 . drwxr-xr-x 3 root root 4096 Sep 18 01:02 .. -rw-r--r-- 1 shalini shalini 40 Sep 19 21:53 .bash_history -rw-r--r-- 1 shalini shalini 220 Feb 13 2026 .bash_logout -rw-r--r-- 1 shalini shalini 3771 Feb 13 2026 .bashrc drwx----- 12 shalini shalini 4096 Sep 19 19:17 .cache drwx----- 15 shalini shalini 4096 Sep 19 21:43 .config drwx----- 4 shalini shalini 4096 Sep 18 01:30 .local -rw-r--r-- 1 shalini shalini 807 Feb 13 2026 .profile drwx----- 2 shalini shalini 4096 Sep 18 01:30 .ssh drwxr-xr-x 2 shalini shalini 4096 Sep 18 01:30 Desktop drwxr-xr-x 2 shalini shalini 4096 Sep 18 01:30 Documents drwxr-xr-x 2 shalini shalini 4096 Sep 18 01:30 Downloads drwxr-xr-x 2 shalini shalini 4096 Sep 18 01:30 Music drwxr-xr-x 2 shalini shalini 4096 Sep 18 01:30 Pictures drwxr-xr-x 2 shalini shalini 4096 Sep 18 01:30 Public drwxr-xr-x 2 shalini shalini 4096 Sep 18 01:30 Templates drwxr-xr-x 2 shalini shalini 4096 Sep 18 01:30 Videos drwx----- 6 shalini shalini 4096 Sep 19 21:00 snap -rw-rw-r-- 1 shalini shalini 28 Sep 20 23:00 testfile -rw-rw-r-- 1 shalini shalini 23 Sep 19 21:53 testfile.save -rw-rw-r-- 1 shalini shalini 28 Sep 21 06:34 testfile3 shalini@shalini-VMware-Virtual-Platform:~\$ </pre>
uname -a , whoami , sudo whoami	 <pre> drwxr-xr-x 2 shalini shalini 4096 Sep 18 01:30 Videos drwx----- 6 shalini shalini 4096 Sep 19 21:00 snap -rw-rw-r-- 1 shalini shalini 28 Sep 20 23:00 testfile -rw-rw-r-- 1 shalini shalini 23 Sep 19 21:53 testfile.save -rw-rw-r-- 1 shalini shalini 28 Sep 21 06:34 testfile3 shalini@shalini-VMware-Virtual-Platform:~\$ mv testfile2 testfile3 mv: cannot stat 'testfile2': No such file or directory shalini@shalini-VMware-Virtual-Platform:~\$ ls -la total 88 drwxr-xr-x 15 shalini shalini 4096 Sep 21 06:36 . drwxr-xr-x 3 root root 4096 Sep 18 01:02 .. -rw-r--r-- 1 shalini shalini 40 Sep 19 21:53 .bash_history -rw-r--r-- 1 shalini shalini 220 Feb 13 2026 .bash_logout -rw-r--r-- 1 shalini shalini 3771 Feb 13 2026 .bashrc drwx----- 12 shalini shalini 4096 Sep 19 19:17 .cache drwx----- 15 shalini shalini 4096 Sep 19 21:43 .config drwx----- 4 shalini shalini 4096 Sep 18 01:30 .local -rw-r--r-- 1 shalini shalini 807 Feb 13 2026 .profile drwx----- 2 shalini shalini 4096 Sep 18 01:30 .ssh drwxr-xr-x 2 shalini shalini 4096 Sep 18 01:30 Desktop drwxr-xr-x 2 shalini shalini 4096 Sep 18 01:30 Documents drwxr-xr-x 2 shalini shalini 4096 Sep 18 01:30 Downloads drwxr-xr-x 2 shalini shalini 4096 Sep 18 01:30 Music drwxr-xr-x 2 shalini shalini 4096 Sep 18 01:30 Pictures drwxr-xr-x 2 shalini shalini 4096 Sep 18 01:30 Public drwxr-xr-x 2 shalini shalini 4096 Sep 18 01:30 Templates drwxr-xr-x 2 shalini shalini 4096 Sep 18 01:30 Videos drwx----- 6 shalini shalini 4096 Sep 19 21:00 snap -rw-rw-r-- 1 shalini shalini 28 Sep 20 23:00 testfile -rw-rw-r-- 1 shalini shalini 23 Sep 19 21:53 testfile.save -rw-rw-r-- 1 shalini shalini 28 Sep 21 06:34 testfile3 shalini@shalini-VMware-Virtual-Platform:~\$ uname -a Linux shalini-VMware-Virtual-Platform 7.0.8-38-generic #30-Ubuntu SMP PREEMPT_DYNAMIC Fri Jul 31 18:22:54 UTC 2026 x86_64 GNU/Linux shalini@shalini-VMware-Virtual-Platform:~\$ whoami shalini shalini@shalini-VMware-Virtual-Platform:~\$ sudo whoami [sudo: authenticate] Password: root shalini@shalini-VMware-Virtual-Platform:~\$ </pre>

More CLI practice: pwd, man, mkdir, ps, top, permissions, find/grep

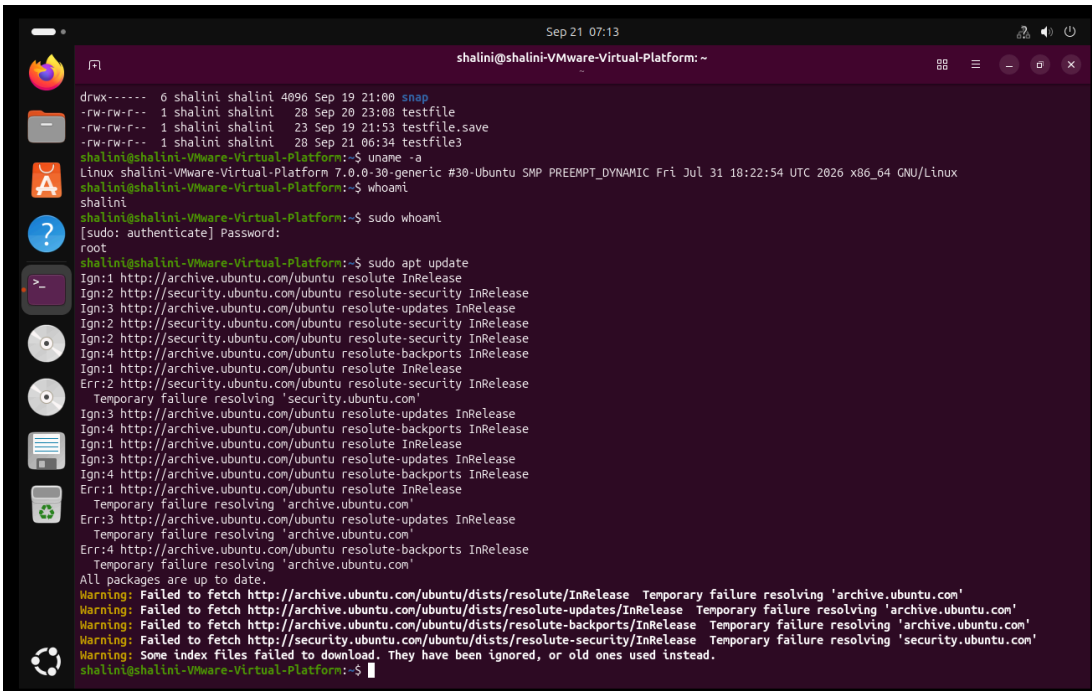
Step	Screenshot
pwd	 <pre>shalini@shalini-VMware-Virtual-Platform: ~ shalini@shalini-VMware-Virtual-Platform:~\$ pwd /home/shalini shalini@shalini-VMware-Virtual-Platform:~\$</pre>
man ls	 <pre>shalini@shalini-VMware-Virtual-Platform: ~ -- man ls LS(1) General Commands Manual LS(1) NAME ls - List directory contents. Ignore files and directories starting with a '.' by default SYNOPSIS ls [--help] [--format] [-C] [-l] [--long] [-x] [-T] [--tabsize] [-n] [--zero] [-D] [--dired] [--hyperlink] [-1] [-o] [-g] [-n] [--numeric-uid-gid] [--quoting-style] [-N] [--literal] [-b] [--escape] [-Q] [--quote-name] [-q] [--hide-control-chars] [--show-control-chars] [--time] [-e] [-w] [--hide] [-Z] [--ignore] [-d] [--ignore-backups] [--sort] [-S] [-t] [-v] [-X] [-u] [-L] [--dereference] [--dereference-command-line-synlink-to-dir] [-H] [--dereference-command-line] [-G] [--no-group] [--author] [-a] [--all] [-A] [--almost-all] [-F] [-d] [--directory] [-h] [--human-readable] [-k] [--kibibytes] [--SI] [--block-size] [--ip-locale] [--reverse] [-R] [--recursive] [-w] [--width] [-s] [--size] [--color] [--indicator-style] [-F] [--classify] [--file-type] [-p] [--time-style] [--full-time] [-Z] [--context] [--group-directories-first] [-v] [--version] [path] DESCRIPTION List directory contents. Ignore files and directories starting with a '.' by default OPTIONS --help Print help information. --format Set the display format. -C Display the files in columns. -l, --long Display detailed information. -x List entries in rows instead of in columns. -T, --tabsize <COLS> Assume tab stops at each COLS instead of 8 Manual page ls(1) line 1/300 22% (press h for help or q to quit)</pre>
mkdir my lab typo made two folders instead of one	 <pre>shalini@shalini-VMware-Virtual-Platform:~\$ mkdir my lab shalini@shalini-VMware-Virtual-Platform:~\$ cd mylab bash: cd: mylab: No such file or directory shalini@shalini-VMware-Virtual-Platform:~\$ ls Desktop Downloads Pictures Templates lab snap testfile.save Documents Music Public Videos my testfile testfile3 shalini@shalini-VMware-Virtual-Platform:~\$</pre>

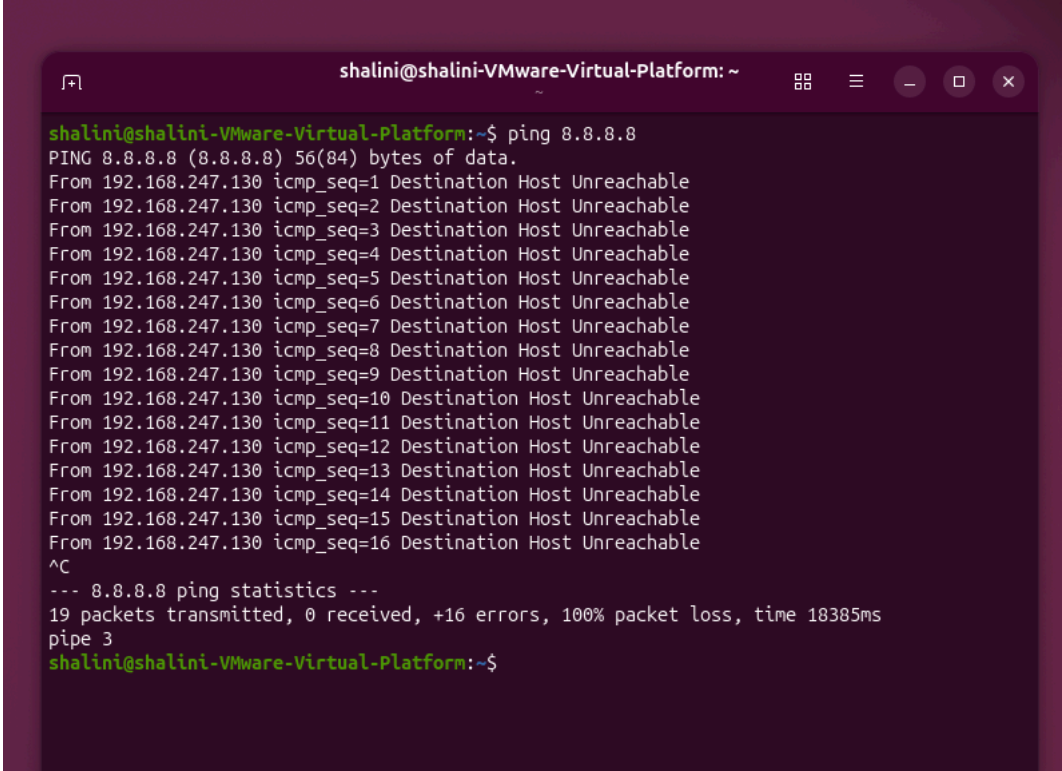
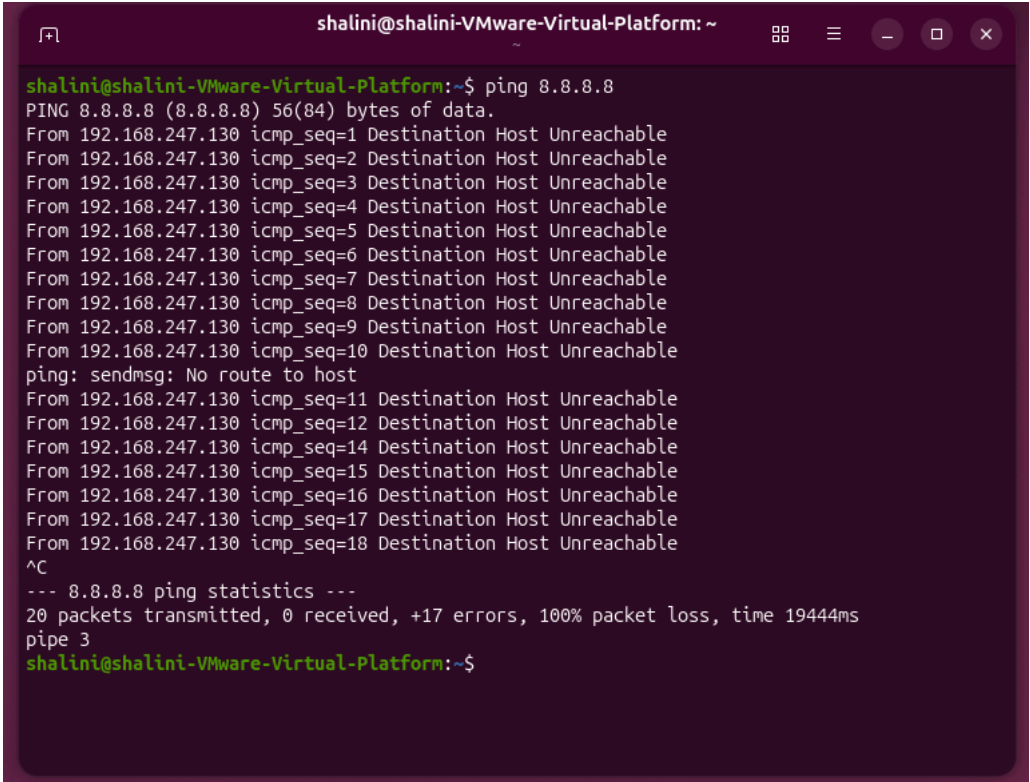
Step	Screenshot
Corrected mkdir mylab	 <pre>shalini@shalini-VMware-Virtual-Platform: ~/lab/mylab shalini@shalini-VMware-Virtual-Platform: ~/lab\$ mkdir mylab shalini@shalini-VMware-Virtual-Platform: ~/lab\$ cd mylab shalini@shalini-VMware-Virtual-Platform: ~/lab/mylab\$ ls shalini@shalini-VMware-Virtual-Platform: ~/lab/mylab\$</pre>
touch file1.txt	 <pre>shalini@shalini-VMware-Virtual-Platform: ~/lab/mylab shalini@shalini-VMware-Virtual-Platform: ~/lab\$ mkdir mylab shalini@shalini-VMware-Virtual-Platform: ~/lab\$ cd mylab shalini@shalini-VMware-Virtual-Platform: ~/lab/mylab\$ ls shalini@shalini-VMware-Virtual-Platform: ~/lab/mylab\$ touch file1.txt shalini@shalini-VMware-Virtual-Platform: ~/lab/mylab\$ ls file1.txt shalini@shalini-VMware-Virtual-Platform: ~/lab/mylab\$</pre>
ps	 <pre>shalini@shalini-VMware-Virtual-Platform: ~/lab/mylab shalini@shalini-VMware-Virtual-Platform: ~/lab\$ mkdir mylab shalini@shalini-VMware-Virtual-Platform: ~/lab\$ cd mylab shalini@shalini-VMware-Virtual-Platform: ~/lab/mylab\$ ls shalini@shalini-VMware-Virtual-Platform: ~/lab/mylab\$ touch file1.txt shalini@shalini-VMware-Virtual-Platform: ~/lab/mylab\$ ls file1.txt shalini@shalini-VMware-Virtual-Platform: ~/lab/mylab\$ ps PID TTY TIME CMD 4112 pts/0 00:00:00 bash 4168 pts/0 00:00:00 ps shalini@shalini-VMware-Virtual-Platform: ~/lab/mylab\$</pre>

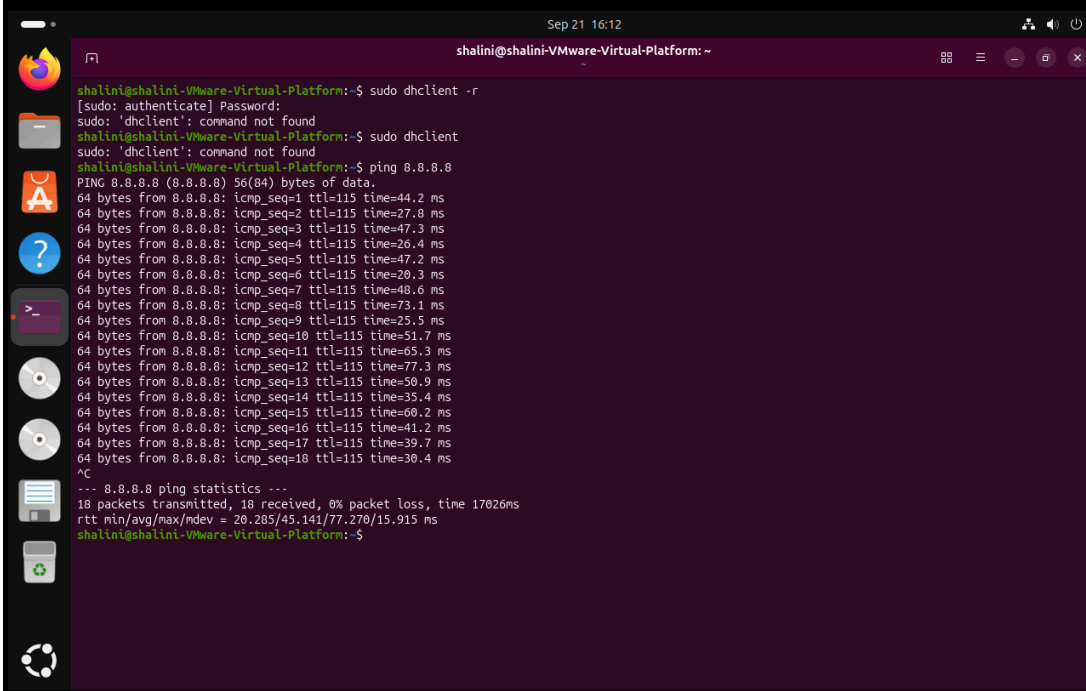
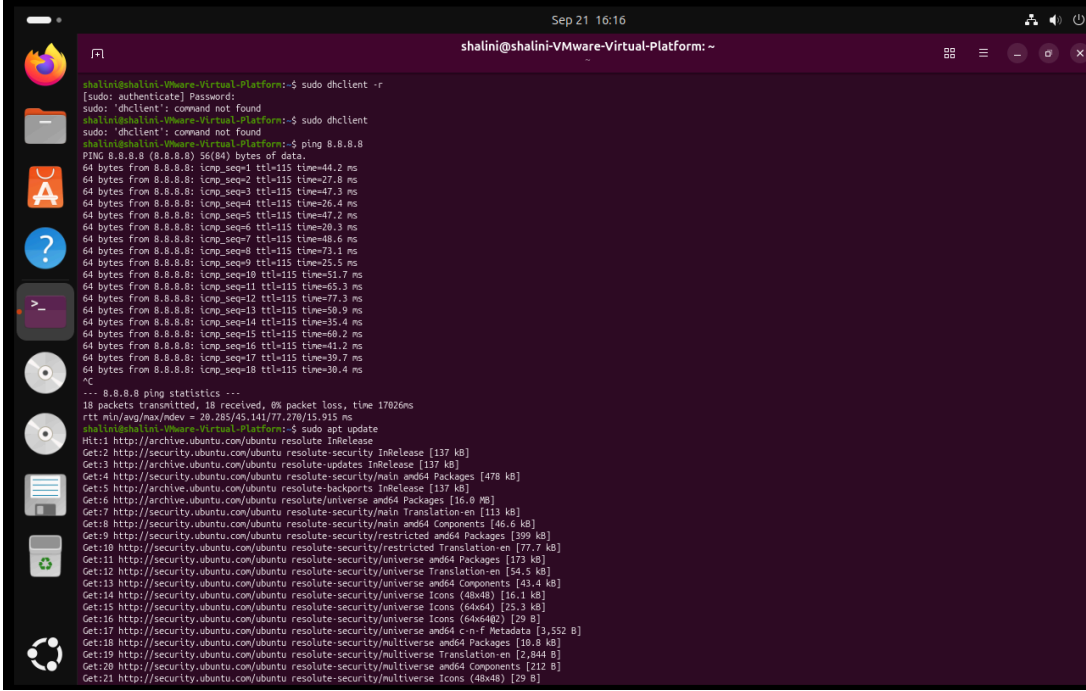
Step	Screenshot
top	
chmod 755 , chown , find	
grep "shalini" /etc/passwd succeeded after two typo attempts (/ect/passwd , /ect/password)	

Troubleshooting: VM lost network access

apt update failed with "Temporary failure resolving", and ping 8.8.8.8 returned "Destination Host Unreachable" / "No route to host". Network connectivity was restored later and apt update then worked.

Step	Screenshot
apt update DNS failure	

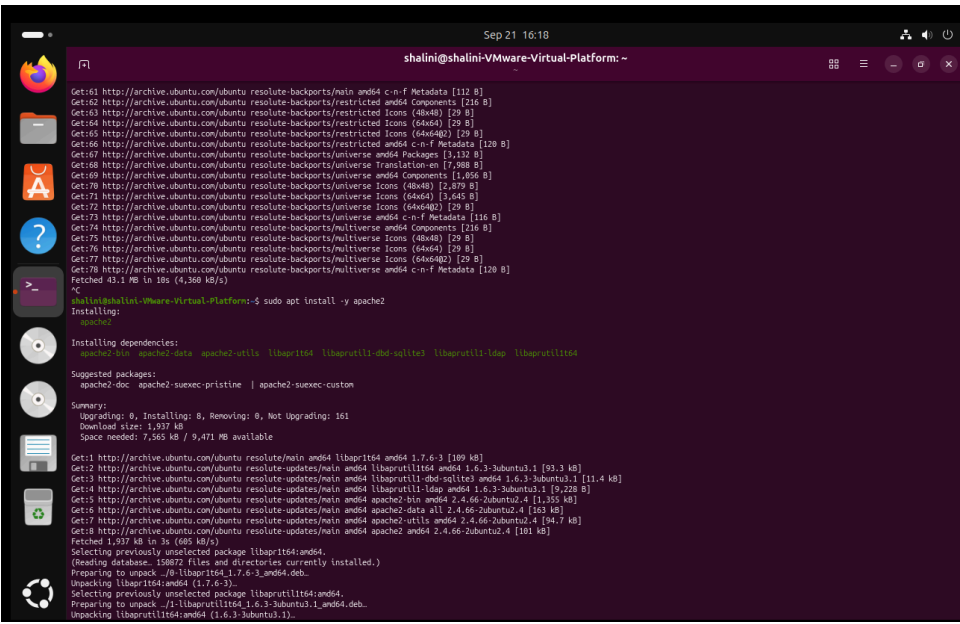
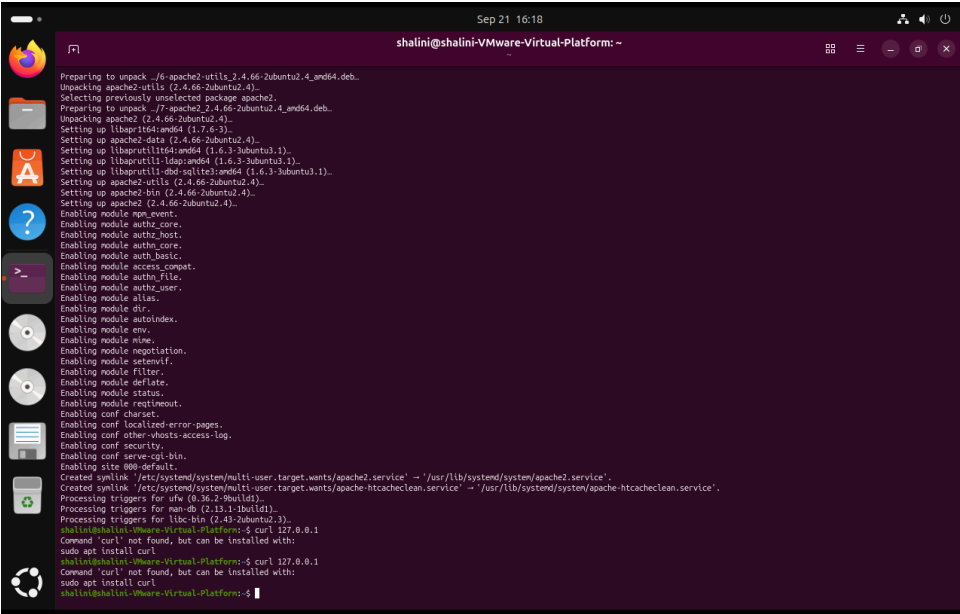
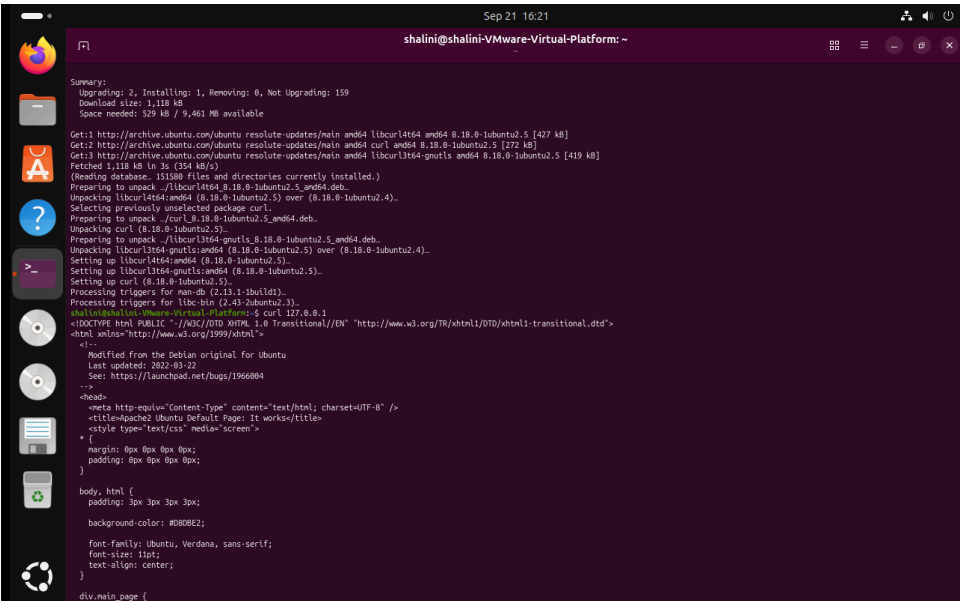
Step	Screenshot
ping unreachable	 <pre> shalini@shalini-VMware-Virtual-Platform: ~ shalini@shalini-VMware-Virtual-Platform:~\$ ping 8.8.8.8 PING 8.8.8.8 (8.8.8.8) 56(84) bytes of data. From 192.168.247.130 icmp_seq=1 Destination Host Unreachable From 192.168.247.130 icmp_seq=2 Destination Host Unreachable From 192.168.247.130 icmp_seq=3 Destination Host Unreachable From 192.168.247.130 icmp_seq=4 Destination Host Unreachable From 192.168.247.130 icmp_seq=5 Destination Host Unreachable From 192.168.247.130 icmp_seq=6 Destination Host Unreachable From 192.168.247.130 icmp_seq=7 Destination Host Unreachable From 192.168.247.130 icmp_seq=8 Destination Host Unreachable From 192.168.247.130 icmp_seq=9 Destination Host Unreachable From 192.168.247.130 icmp_seq=10 Destination Host Unreachable From 192.168.247.130 icmp_seq=11 Destination Host Unreachable From 192.168.247.130 icmp_seq=12 Destination Host Unreachable From 192.168.247.130 icmp_seq=13 Destination Host Unreachable From 192.168.247.130 icmp_seq=14 Destination Host Unreachable From 192.168.247.130 icmp_seq=15 Destination Host Unreachable From 192.168.247.130 icmp_seq=16 Destination Host Unreachable ^C --- 8.8.8.8 ping statistics --- 19 packets transmitted, 0 received, +16 errors, 100% packet loss, time 18385ms pipe 3 shalini@shalini-VMware-Virtual-Platform:~\$ </pre>
ping no route to host	 <pre> shalini@shalini-VMware-Virtual-Platform: ~ shalini@shalini-VMware-Virtual-Platform:~\$ ping 8.8.8.8 PING 8.8.8.8 (8.8.8.8) 56(84) bytes of data. From 192.168.247.130 icmp_seq=1 Destination Host Unreachable From 192.168.247.130 icmp_seq=2 Destination Host Unreachable From 192.168.247.130 icmp_seq=3 Destination Host Unreachable From 192.168.247.130 icmp_seq=4 Destination Host Unreachable From 192.168.247.130 icmp_seq=5 Destination Host Unreachable From 192.168.247.130 icmp_seq=6 Destination Host Unreachable From 192.168.247.130 icmp_seq=7 Destination Host Unreachable From 192.168.247.130 icmp_seq=8 Destination Host Unreachable From 192.168.247.130 icmp_seq=9 Destination Host Unreachable From 192.168.247.130 icmp_seq=10 Destination Host Unreachable ping: sendmsg: No route to host From 192.168.247.130 icmp_seq=11 Destination Host Unreachable From 192.168.247.130 icmp_seq=12 Destination Host Unreachable From 192.168.247.130 icmp_seq=14 Destination Host Unreachable From 192.168.247.130 icmp_seq=15 Destination Host Unreachable From 192.168.247.130 icmp_seq=16 Destination Host Unreachable From 192.168.247.130 icmp_seq=17 Destination Host Unreachable From 192.168.247.130 icmp_seq=18 Destination Host Unreachable ^C --- 8.8.8.8 ping statistics --- 20 packets transmitted, 0 received, +17 errors, 100% packet loss, time 19444ms pipe 3 shalini@shalini-VMware-Virtual-Platform:~\$ </pre>

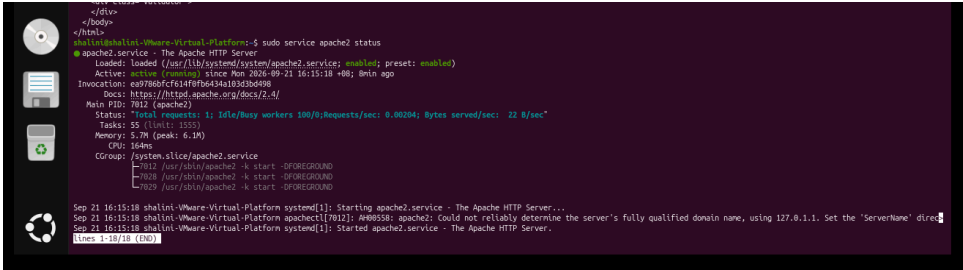
Step	Screenshot
ping restored (0% loss)	 <pre> shalini@shalini-VMware-Virtual-Platform:~\$ sudo dhclient -r [sudo: authenticate] Password: sudo: 'dhclient': command not found shalini@shalini-VMware-Virtual-Platform:~\$ sudo dhclient sudo: 'dhclient': command not found shalini@shalini-VMware-Virtual-Platform:~\$ ping 8.8.8.8 PING 8.8.8.8 (8.8.8.8) 56(84) bytes of data: 64 bytes from 8.8.8.8: icmp_seq=1 ttl=115 time=44.2 ms 64 bytes from 8.8.8.8: icmp_seq=2 ttl=115 time=27.8 ms 64 bytes from 8.8.8.8: icmp_seq=3 ttl=115 time=47.3 ms 64 bytes from 8.8.8.8: icmp_seq=4 ttl=115 time=26.4 ms 64 bytes from 8.8.8.8: icmp_seq=5 ttl=115 time=47.2 ms 64 bytes from 8.8.8.8: icmp_seq=6 ttl=115 time=20.3 ms 64 bytes from 8.8.8.8: icmp_seq=7 ttl=115 time=48.6 ms 64 bytes from 8.8.8.8: icmp_seq=8 ttl=115 time=73.1 ms 64 bytes from 8.8.8.8: icmp_seq=9 ttl=115 time=25.5 ms 64 bytes from 8.8.8.8: icmp_seq=10 ttl=115 time=51.7 ms 64 bytes from 8.8.8.8: icmp_seq=11 ttl=115 time=65.3 ms 64 bytes from 8.8.8.8: icmp_seq=12 ttl=115 time=77.3 ms 64 bytes from 8.8.8.8: icmp_seq=13 ttl=115 time=50.9 ms 64 bytes from 8.8.8.8: icmp_seq=14 ttl=115 time=35.4 ms 64 bytes from 8.8.8.8: icmp_seq=15 ttl=115 time=60.2 ms 64 bytes from 8.8.8.8: icmp_seq=16 ttl=115 time=41.2 ms 64 bytes from 8.8.8.8: icmp_seq=17 ttl=115 time=39.7 ms 64 bytes from 8.8.8.8: icmp_seq=18 ttl=115 time=30.4 ms ^C --- 8.8.8.8 ping statistics --- 18 packets transmitted, 18 received, 0% packet loss, time 17026ms rtt min/avg/max/mdev = 20.285/45.141/77.270/15.915 ms shalini@shalini-VMware-Virtual-Platform:~\$ </pre>
apt update working	 <pre> shalini@shalini-VMware-Virtual-Platform:~\$ sudo dhclient -r [sudo: authenticate] Password: sudo: 'dhclient': command not found shalini@shalini-VMware-Virtual-Platform:~\$ sudo dhclient sudo: 'dhclient': command not found shalini@shalini-VMware-Virtual-Platform:~\$ ping 8.8.8.8 PING 8.8.8.8 (8.8.8.8) 56(84) bytes of data: 64 bytes from 8.8.8.8: icmp_seq=1 ttl=115 time=44.2 ms 64 bytes from 8.8.8.8: icmp_seq=2 ttl=115 time=27.8 ms 64 bytes from 8.8.8.8: icmp_seq=3 ttl=115 time=47.3 ms 64 bytes from 8.8.8.8: icmp_seq=4 ttl=115 time=26.4 ms 64 bytes from 8.8.8.8: icmp_seq=5 ttl=115 time=47.2 ms 64 bytes from 8.8.8.8: icmp_seq=6 ttl=115 time=20.3 ms 64 bytes from 8.8.8.8: icmp_seq=7 ttl=115 time=48.6 ms 64 bytes from 8.8.8.8: icmp_seq=8 ttl=115 time=73.1 ms 64 bytes from 8.8.8.8: icmp_seq=9 ttl=115 time=25.5 ms 64 bytes from 8.8.8.8: icmp_seq=10 ttl=115 time=51.7 ms 64 bytes from 8.8.8.8: icmp_seq=11 ttl=115 time=65.3 ms 64 bytes from 8.8.8.8: icmp_seq=12 ttl=115 time=77.3 ms 64 bytes from 8.8.8.8: icmp_seq=13 ttl=115 time=50.9 ms 64 bytes from 8.8.8.8: icmp_seq=14 ttl=115 time=35.4 ms 64 bytes from 8.8.8.8: icmp_seq=15 ttl=115 time=60.2 ms 64 bytes from 8.8.8.8: icmp_seq=16 ttl=115 time=41.2 ms 64 bytes from 8.8.8.8: icmp_seq=17 ttl=115 time=39.7 ms 64 bytes from 8.8.8.8: icmp_seq=18 ttl=115 time=30.4 ms ^C --- 8.8.8.8 ping statistics --- 18 packets transmitted, 18 received, 0% packet loss, time 17026ms rtt min/avg/max/mdev = 20.285/45.141/77.270/15.915 ms shalini@shalini-VMware-Virtual-Platform:~\$ sudo apt update Hit:1 http://archive.ubuntu.com/ubuntu resolute InRelease Get:2 http://security.ubuntu.com/ubuntu resolute-security InRelease [137 kB] Get:3 http://archive.ubuntu.com/ubuntu resolute-updates InRelease [137 kB] Get:4 http://security.ubuntu.com/ubuntu resolute-security/main amd64 Packages [478 kB] Get:5 http://archive.ubuntu.com/ubuntu resolute-backports InRelease [137 kB] Get:6 http://archive.ubuntu.com/ubuntu resolute/universe amd64 Packages [16.0 MB] Get:7 http://security.ubuntu.com/ubuntu resolute-security/main Translation-en [113 kB] Get:8 http://security.ubuntu.com/ubuntu resolute-security/main amd64 Components [46.6 kB] Get:9 http://security.ubuntu.com/ubuntu resolute-security/restricted amd64 Packages [399 kB] Get:10 http://security.ubuntu.com/ubuntu resolute-security/restricted Translation-en [77.7 kB] Get:11 http://security.ubuntu.com/ubuntu resolute-security/universe amd64 Packages [173 kB] Get:12 http://security.ubuntu.com/ubuntu resolute-security/universe Translation-en [54.5 kB] Get:13 http://security.ubuntu.com/ubuntu resolute-security/universe amd64 Components [43.4 kB] Get:14 http://security.ubuntu.com/ubuntu resolute-security/universe Icons (48x48) [16.1 kB] Get:15 http://security.ubuntu.com/ubuntu resolute-security/universe Icons (64x64) [25.3 kB] Get:16 http://security.ubuntu.com/ubuntu resolute-security/universe Icons (64x64@2) [29 B] Get:17 http://security.ubuntu.com/ubuntu resolute-security/universe amd64 c-n-f Metadata [3,552 B] Get:18 http://security.ubuntu.com/ubuntu resolute-security/multiverse amd64 Packages [18.8 kB] Get:19 http://security.ubuntu.com/ubuntu resolute-security/multiverse Translation-en [2,844 B] Get:20 http://security.ubuntu.com/ubuntu resolute-security/multiverse amd64 Components [212 B] Get:21 http://security.ubuntu.com/ubuntu resolute-security/multiverse Icons (48x48) [29 B] </pre>

Fix: [add what you did to restore the network, e.g. VMware network adapter setting / reconnect]

Lab 1b – Linux services: Apache

Commands: `sudo apt install -y apache2`, `sudo apt install curl`, `curl 127.0.0.1`, `sudo service apache2 status`

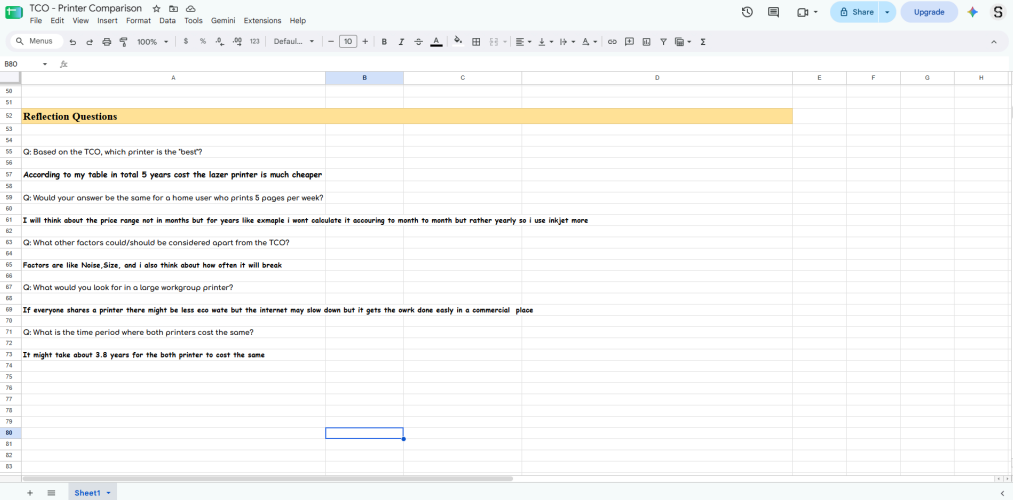
Step	Screenshot
Install Apache	
Install complete; curl was missing	
Installed curl, "Apache2 Ubuntu Default Page: It works"	

Step	Screenshot
Service active (running)	

The status log shows a harmless warning that the server's fully qualified domain name could not be determined, so it defaulted to 127.0.1.1. Setting `ServerName` in the Apache config removes it.

Lab 2a – Total Cost of Ownership (TCO) Comparison

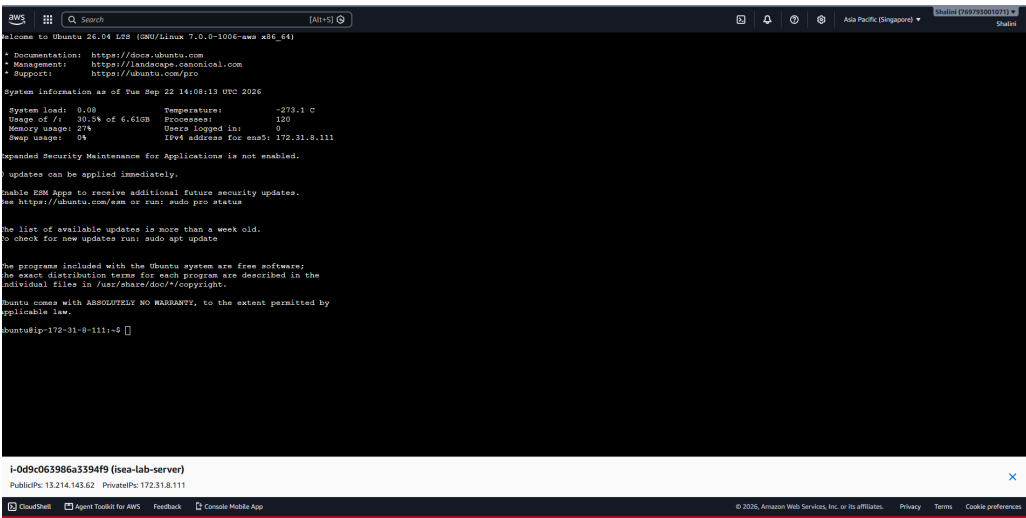
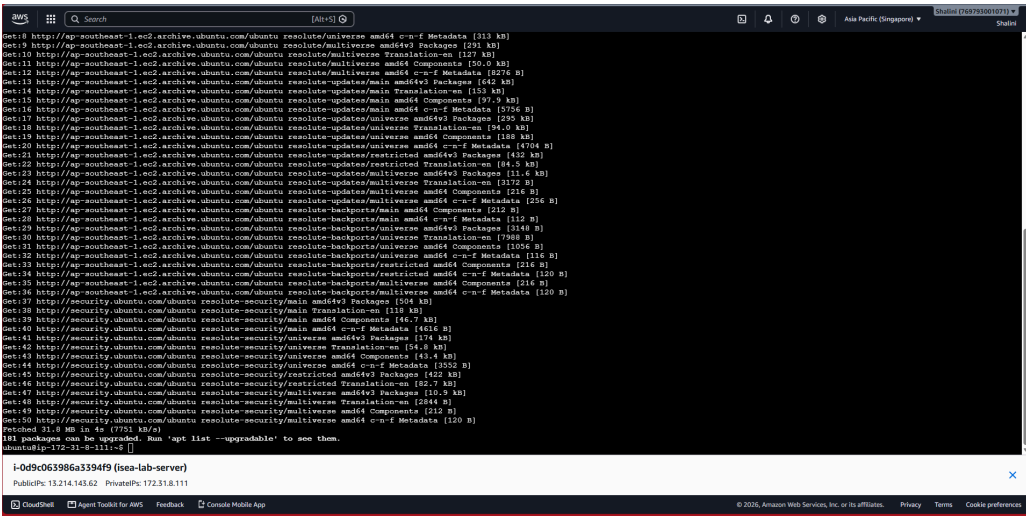
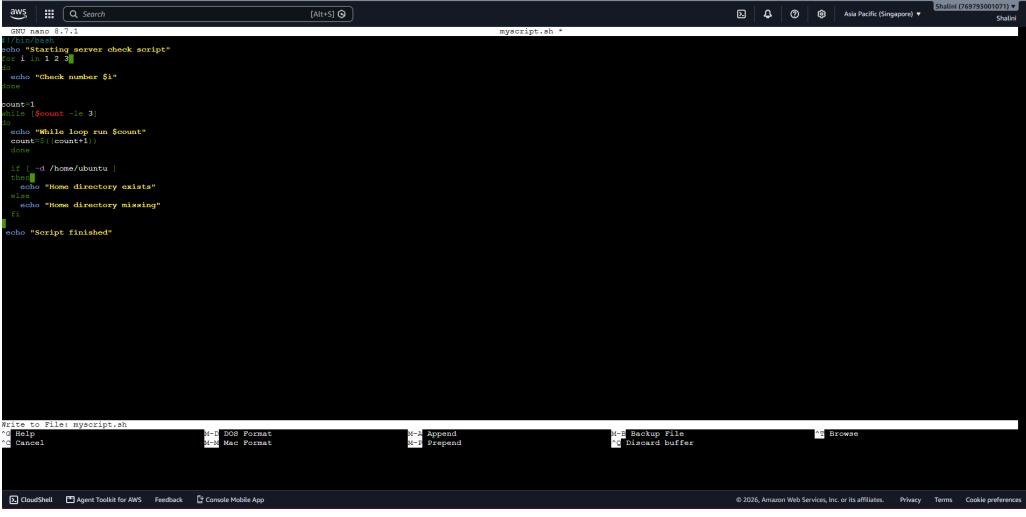
Compared a budget inkjet printer against an entry-level laser printer over 5 years, using the lab's default assumptions (750 pages/week, 40 hours/week powered on). Built as an Excel spreadsheet with linked formulas, not hardcoded numbers.

Step	Screenshot
TCO comparison spreadsheet with reflection answers	

Result: the laser printer is far cheaper over the 5-year period, despite its higher purchase price, because its running cost per page is much low — break-even is reached in under 4 weeks at this print volume.

Lab 2b – Cloud VM (AWS EC2) + Bash scripting

Launched a free-tier Ubuntu EC2 instance (t3.micro) in the Asia Pacific (Singapore) region, connected via browser-based SSH (EC2 Instance Connect), updated packages, then wrote and ran a bash script covering `echo`, `for`, `while`, and `if`.

Step	Screenshot
Connected to the EC2 instance	 The screenshot shows the AWS CloudShell interface with a terminal window. The terminal output displays the Ubuntu 26.04 LTS (GNU/Linux 7.0.0-1006-aws x86_64) welcome message, documentation links, and system information as of Tue Sep 22 14:08:13 UTC 2026. It shows system load, memory usage, and network status. A message indicates that expanded security maintenance for applications is not enabled and updates can be applied immediately. It also mentions the availability of updates and the need to check for new updates. The terminal ends with the prompt 'ubuntu@ip-172-31-8-111:~\$'.
sudo apt update	 The screenshot shows the AWS CloudShell interface with a terminal window. The terminal output displays the result of running 'sudo apt update'. It lists various packages and their versions, including 'resoluto/universe amd64 c-n-f Metadata', 'resoluto/multiverse amd64 Packages', 'resoluto/multiverse amd64 Components', 'resoluto-updates/main amd64 Packages', 'resoluto-updates/main amd64 Components', 'resoluto-updates/universe amd64 Components', 'resoluto-updates/universe amd64 c-n-f Metadata', 'resoluto-updates/restricted amd64 Packages', 'resoluto-updates/restricted Translation-en', 'resoluto-updates/multiverse amd64 Packages', 'resoluto-updates/multiverse amd64 Components', 'resoluto-updates/multiverse amd64 c-n-f Metadata', 'resoluto-backports/main amd64 Components', 'resoluto-backports/main amd64 c-n-f Metadata', 'resoluto-backports/restricted amd64 Packages', 'resoluto-backports/restricted Translation-en', 'resoluto-backports/universe amd64 Packages', 'resoluto-backports/universe amd64 c-n-f Metadata', 'resoluto-backports/universe amd64 Components', 'resoluto-security/main amd64 Components', 'resoluto-security/main amd64 c-n-f Metadata', 'resoluto-security/universe amd64 Packages', 'resoluto-security/universe amd64 Components', 'resoluto-security/universe amd64 c-n-f Metadata', 'resoluto-security/restricted amd64 Packages', 'resoluto-security/restricted Translation-en', 'resoluto-security/multiverse amd64 Packages', 'resoluto-security/multiverse amd64 Components', and 'resoluto-security/multiverse amd64 c-n-f Metadata'. It also shows the total size of the updates and the prompt 'ubuntu@ip-172-31-8-111:~\$'.
Wrote myscript.sh in nano	 The screenshot shows the AWS CloudShell interface with a terminal window. The terminal output displays the content of a file named 'myscript.sh'. The script starts with a shebang line '#!/bin/bash' and a comment '# Starting server check script'. It then defines a function 'check_number' that takes a number as an argument and checks if it is greater than 1. If it is, it prints 'Home directory exists'. If it is not, it prints 'Home directory missing'. The script ends with a comment '# Script finished'.
	 The screenshot shows the AWS CloudShell interface with a terminal window. The terminal output displays the Ubuntu 26.04 LTS (GNU/Linux 7.0.0-1006-aws x86_64) welcome message, documentation links, and system information as of Tue Sep 22 14:13:41 UTC 2026. It shows system load, memory usage, and network status. A message indicates that expanded security maintenance for applications is not enabled.

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